Checking and securing understanding of non-linear sequences

1 The sequence -7, -3, 1, 5, 9, ... has a common difference of +4 therefore it is called ___________ sequence. Tick **1** correct answer

- [x] an arithmetic
- [ ] a counting
- [ ] a geometric

2 The sequence 4, 12, 36, 108, ...has a common ratio of ×4 therefore it is called ___________ sequence. Tick **1** correct answer

- [ ] an arithmetic
- [x] a geometric
- [ ] a multiplier
- [ ] a multiplier

3 Match each arithmetic sequence to its next term. Write the correct letter in each box

a	3, 5, 7, 9, 11, ...
b	27, 24, 21, 18, 15, ...
c	-19, -13, -7, -1, 5, ...
d	-10, -6, -2, 2, 6, ...

b	12
d	10
c	11
a	13

4 The next term of this geometric sequence 5, 15, 45, 135, ... is 405 . Fill in the blank

5 Why is 50, 45, 39, 32, 44, ... not an arithmetic sequence? Tick **1** correct answer

- [ ] Arithmetic sequences can not decrease.
- [ ] It is an arithmetic sequence. The differences go -5, -6, -7, -8, ...
- [x] It does not have a common difference between terms.

6 Which of these words describes the sequence -86, -74, -62, -50, … ? Tick **3** correct answers

- [x] Additive
- [x] Arithmetic
- [ ] Decreasing
- [ ] Geometric
- [x] Linear